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Temperature Effects on Battery Dispatch for Photovoltaic Curtailment Recovery Under Export Constraints

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Abstract: Battery energy storage can improve the utilization of available photovoltaic (PV) generation by recovering energy otherwise curtailed under grid export limits, but simplified dispatch models can overlook temperature-dependent changes in usable capacity and power capability. This study investigates how environmental temperature and current-driven self-heating affect curtailment recovery and the adequacy of simplified battery models. Public five-minute weather data for Phoenix were combined with a reduced lithium iron phosphate (LFP) battery model derived from a dynamic electrothermal model independently validated against held-out profiles at 25 and 35 ℃. Annual dispatch simulations compared electrical-only optimization, terminal-state-of-charge-corrected nonlinear replay, and coupled sequential-linearization dispatch. For a 230 MW PV plant with a 195.5 MW export limit and a 30 MW/140 MWh battery, annual curtailment decreased from 39.601 GWh without storage to 8.727 GWh under electrical-only optimization. Nonlinear replay with terminal correction yielded 9.032 GWh, while coupled dispatch reduced this value to 8.956 GWh. On a common 112 MWh usable-energy basis defined at 25 ℃, ambient-driven variation in electrical properties increased annual curtailment by 265.6 MWh relative to a fixed-25 ℃ reference under an exposed-cell thermal boundary. Temperature-dependent usable capacity accounted for most of this difference; alternative empirical capacity maps yielded effects of 233.4–295.5 MWh/year. Current-driven self-heating produced no numerically resolvable annual curtailment effect at the nominal rating. Extending the dispatch horizon from 24 to 48 hours changed annual coupled-dispatch curtailment by 0.0125%. These findings show that weak self-heating alone does not establish the adequacy of a simplified battery model. Comparisons on a common reference usable-energy basis, with consistent terminal state of charge (SOC) conditions and explicit physical-feasibility checks, provide a basis for selecting battery models to assess renewable energy recovery in export-constrained PV-storage systems.

Keywords: Photovoltaic curtailment; Battery energy storage; Electrothermal modeling; Temperature-dependent capacity; Export-constrained dispatch; Model adequacy

1 Introduction

Grid export limits restrict the amount of photovoltaic (PV) generation that a plant can deliver through its point of interconnection (POI). Although these limits help accommodate renewable generation within network constraints, they can leave part of the available solar resource unused [1, 2]. Battery energy storage systems (BESSs) can recover this energy by charging when PV output exceeds the export limit and discharging when export headroom becomes available. The amount recovered depends on both storage configuration and dispatch decisions. Previous studies have examined PV–battery plant configuration [3], storage sizing for clipping-loss recovery [4], multi-service operation with battery ageing [5], and the effects of temporal resolution and nonlinear battery dynamics [6]. The value of colocating renewable generation and storage also depends on interconnection and operational conditions [7]. These considerations make dispatch modeling central to the assessment of export-constrained PV–storage systems, for which optimization and model predictive control are established energy-management tools [8].

System-level dispatch models commonly represent a battery using state of charge (SOC), fixed conversion efficiencies, a prescribed energy capacity, and constant power limits. This formulation supports efficient calculations

over long operating periods, but it does not explicitly describe changes in voltage, resistance, usable capacity, or power capability with temperature. Electrothermal modeling provides a physical basis for representing these changes by linking electrical behavior to heat generation and heat transfer [9]. Coupled models have been applied to battery charging control [10], dynamic state-of-power estimation [11], temperature-dependent hybrid-system optimization [12], module-level scheduling [13], and ageing-aware predictive control [14]. Temperature-dependent equivalent-circuit models have also been developed for lithium iron phosphate (LFP) cells [15]. Together, these studies establish the relevance of temperature to battery operation; its importance for a particular dispatch objective depends on how the resulting electrical changes affect the operating trajectory.

For PV curtailment recovery, two temperature-related mechanisms require separate assessment. First, the external thermal environment changes cell temperature and therefore the electrical properties that govern charge storage, energy conversion, and feasible power. Second, battery current generates heat, which changes temperature and feeds back on those same properties. These mechanisms are related, but their effects on recovered energy need not be comparable. Weak self-heating does not remove the influence of ambient-driven changes in usable capacity. Conversely, a measurable temperature rise does not necessarily change curtailment if the battery remains within its operating limits. A dispatch-level assessment must therefore connect the thermal response to stored charge, accepted power, and recovered PV energy.

1.1 Related Work and Research Gap

Table 1 places the present study alongside relevant PV–battery and electrothermal modeling research. Plant-level studies provide methods for evaluating storage configuration, clipping recovery, and dispatch, while cell- and module-level studies describe temperature-dependent electrical behavior and operating constraints. Hybrid-system optimization connects these two scales by incorporating temperature into energy-management decisions. The question addressed here is how the external thermal environment and current-driven self-heating contribute separately to curtailment recovery under a hard POI export limit.

Table 1. Related studies and the focus of the present work

Study	Application and Modeling Focus	Relation to the Present Study
DiOrio et al. [3]	PV–battery configuration and dispatch	Plant-level framework for evaluating storage operation and recovery of otherwise unused PV generation
Bahloul et al. [4]	Storage sizing for PV clipping-loss recovery	Direct connection between storage configuration and recoverable PV energy
Bahloul et al. [5]	Multi-service dispatch of colocated PV and storage with battery ageing	Dispatch under competing operational objectives
Castillejo-Cuberos et al. [6]	High-resolution PV–battery techno-economic modeling	Effects of temporal resolution and nonlinear battery dynamics
de Souza et al. [10]	Charging control using a coupled electrothermal model	Cell-level representation of electrical and thermal feedback
Xavier et al. [11]	State-of-power estimation using an electrothermal model	Temperature-dependent battery power feasibility
Mishra et al. [12]	Predictive optimization of temperature-dependent hybrid energy systems	Integration of temperature dependence into system-level energy management
Wu et al. [13]	BESS scheduling with module temperature differences	Connection between thermal conditions and storage scheduling
Dorronsoro et al. [14]	Predictive control using coupled electrical, thermal, and ageing models	Control decisions informed by interacting battery states
Present study	Temperature-dependent dispatch for PV curtailment recovery under a hard export limit	Matched-energy model comparisons, separate attribution of ambient-property and self-heating effects, and physical-feasibility assessment

Note: PV = photovoltaic; BESS = battery energy storage system.

This distinction also affects comparisons between simplified and coupled battery models. A difference in predicted curtailment can arise from unequal usable-energy assumptions, inconsistent terminal SOC, temperature-dependent state conversion, or physical limits imposed during schedule execution. Comparing objective values alone does not separate these causes. A useful assessment therefore requires a common reference usable-energy basis,

consistent terminal-SOC conditions, and nonlinear replay of the requested battery power. Controlled comparisons can then distinguish the effect of ambient-driven electrical-property variation from the additional feedback caused by self-heating.

1.2 Research Objectives and Contributions

This study examines how temperature affects battery dispatch and curtailment recovery in an export-constrained PV plant. It addresses three questions: how much ambient-driven electrical-property variation and current-driven self-heating each contribute to annual curtailment; how the thermal boundary and battery power-to-energy ratio affect self-heating and physical feasibility; and which differences matter when selecting a simplified battery representation.

The analysis combines a public five-minute solar and weather record for Phoenix with an experimentally parameterized A123 ANR26650M1-B LFP cell model. Dynamic profiles at 25 and 35 ℃ from the public battery dataset are reserved for independent validation [16]. Four operating cases are evaluated: no storage (B0), electrical-only optimization (B1), nonlinear physical replay of the B1 schedule with terminal-SOC correction (B2), and coupled dispatch using sequential linearization (B3). The storage comparisons use a common usable-energy basis defined at 25 ℃ and consistent terminal-SOC conditions.

The study makes three contributions. First, the B1–B2–B3 comparison separates the performance predicted by a simplified dispatch model from the physically accepted operation of its schedule and the recovery obtained through re-optimization with the nonlinear battery representation. Second, controlled B3 comparisons distinguish ambient-driven electrical-property effects from current-driven self-heating, while property-level ablations identify the main source of the annual curtailment difference. Third, a temperature–power-to-energy-ratio map assesses thermal response and command feasibility within the identified parameter domain. Tests of temporal resolution, dispatch horizon, initialization, and parameter sensitivity establish the numerical scale at which these differences can be interpreted.

These comparisons provide a basis for choosing battery-model detail according to its effect on recoverable PV energy and feasible operation. They connect electrothermal behavior to a specific renewable-energy utilization problem while maintaining a common reference-energy basis and consistent terminal-SOC conditions across models.

2 System Model and Dispatch Formulation

Figure 1 shows the PV plant and BESS connected to the grid through a shared, export-limited POI, together with the feedback between battery dispatch and the electrothermal cell model. The battery charges only from PV power above the export limit and discharges only when export headroom is available. This operating rule isolates the recovery of otherwise curtailed generation. The battery model links accepted power to cell current, SOC, and temperature. Nonlinear replay evaluates the physical execution of a prescribed schedule, while coupled dispatch uses the resulting state-conversion coefficients and power limits to update that schedule.

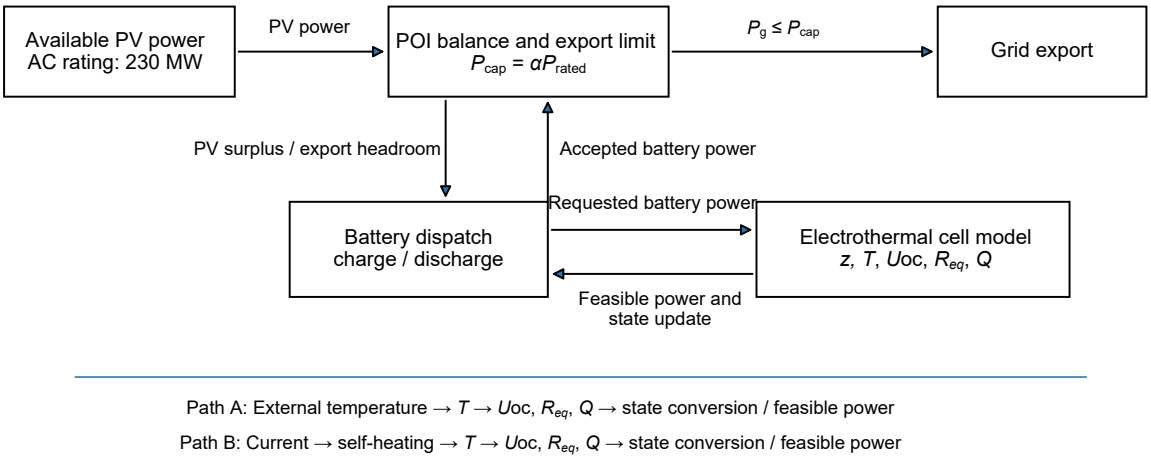


Figure 1. PV–battery system boundary and electrothermal feedback under a POI export limit

Note: PV = photovoltaic; POI = point of interconnection; AC = alternating current; P_{rated} = the plant rating; α = the export-cap ratio; P_{cap} = POI limit; z = state of charge; T = temperature; U_{oc} = open-circuit voltage; R_{eq} = quasi-steady resistance; Q = capacity; P_{g} = grid export power.

2.1 Photovoltaic Plant and Export Constraint

The reference plant has an alternating current (AC) rating of 230 MW. Let $P_{\text{pv},k}$ denote available AC PV power before POI curtailment at interval k, P_{rated} is the plant rating, and α is the export-cap ratio. The POI limit is:

$$P_{\text{cap}} = \alpha P_{\text{rated}} \tag{1}$$

For the nominal case, α = 0.85, giving P_{cap} = 195.5 MW. Available surplus and export headroom are defined as:

$$\begin{aligned} \mid s_k \mid &= \max\left(P_{\text{pv},k} - P_{\text{cap},0}\right) \\ \mid h_k \mid &= \max\left(P_{\text{cap}} - P_{\text{pv},k,0}\right) \end{aligned} \tag{2}$$

Battery charging and discharging powers, $P_{\text{ch},k}$ and $P_{\text{dis},k}$ are nonnegative AC-side quantities constrained by:

$$\begin{aligned} 0 \le P_{\text{ch},k} &\le \min\left(P_{\max}, s_k\right) \\ 0 \le P_{\text{dis},k} &\le \min\left(P_{\max}, h_k\right) \end{aligned} \tag{3}$$

where, P_{max} is the BESS AC power rating. Since surplus and headroom cannot both be positive, these constraints exclude simultaneous charging and discharging. Curtailed power and grid export are:

$$\begin{aligned} \mid P_{\text{curt},k} \mid &= s_k - P_{\text{ch},k}, \\ \mid P_{\text{g},k} \mid &= P_{\text{pv},k} - P_{\text{curt},k} - P_{\text{ch},k} + P_{\text{dis},k} \le P_{\text{cap}} \end{aligned} \tag{4}$$

The same POI constraints apply to all dispatch cases. During nonlinear replay, the charging and discharging powers in Eq. (4) are the physically accepted values after battery constraints have been applied.

2.2 Electrical Battery Model and Energy Normalization

The representative-cell model uses SOC z, temperature T, open-circuit voltage (OCV, $U_{\text{oc}}(z,T)$), ohmic resistance $R_0(z,T)$, and capacity $Q(T)$. A first-order polarization branch was identified from relaxation data. Its characteristic time constant ranged from approximately 49 to 70 s across the identified temperature domain, compared with the 900 s interval used for annual dispatch. The dispatch model therefore adopts the quasi-steady resistance:

$$R_{\text{eq}}(z,T) = R_0(z,T) + R_1(T) \tag{5}$$

where, $R_1(T)$ is the polarization resistance. The full polarization state is retained for the held-out dynamic validation described in Section 3.3.

Cell current I is positive during charging and negative during discharge. Terminal voltage is calculated as $V = U_{\text{oc}}(z,T) + IR_{\text{eq}}(z,T)$.

The discrete SOC update is:

$$z_{k+1} = z_k + \frac{I_k \Delta t}{Q\left(T_k\right)} \tag{6}$$

where, I_k is expressed in amperes, $Q(T_k)$ in ampere-hours, and Δt in hours. Electrical properties are evaluated at the beginning of each interval. Temperature therefore affects the SOC increment through the capacity map, as well as the current–voltage relation through open-circuit voltage and resistance.

The adopted current limits are 10 A during charging and 50 A during discharge. Terminal voltage is restricted to 2.0–3.6 V, and the dispatch SOC range is 0.10–0.90. The charging bound follows the manufacturer's recommended fast-charge current. The continuous discharge rating and charge-voltage endpoint are taken from the 2012 A123 technical data, while the lower voltage bound follows the endpoint shown in its discharge-characteristic data [17]. The SOC window is a study operating assumption.

An energy-equivalent population of identical cells connects the cell model to the plant-level BESS. Its size is normalized at 25 ℃ over the dispatch SOC window:

$$N_{\text{eq}} = \frac{10^6 E_{\text{use}}}{Q_{25} \int_{0.10}^{0.90} U_{\text{oc}}\left(z, 25^{\circ}\text{C}\right) \mathrm{d}z} \tag{7}$$

where, E_{use} is the reference usable energy in MWh and Q_{25} is the identified cell capacity at 25 °C. For the nominal 140 MWh BESS, Euse = 112 MWh. The identified capacity of 2.580 Ah and OCV integral of 2.634 V give approximately 6.796 Wh per equivalent cell over the operating window. The resulting equivalent population is approximately 1.648

$\times 10^7$ cells. Unrounded parameter values are used in the calculation to match the 112 MWh usable-energy window represented in the electrical-only model.

The corresponding OCV-integrated energy over the full SOC range is 139.1 MWh. Normalization is imposed over the operating window because this is the energy available to dispatch. The equivalent population provides an aggregate scaling basis without specifying a commercial pack topology.

Power-conversion-system (PCS) efficiency is 0.98 in each direction. During charging, aggregate cell-terminal power is the accepted AC charging power multiplied by PCS efficiency. During discharge, the required cell-terminal output is the accepted AC discharge power divided by PCS efficiency. Representative-cell current is obtained from the resulting terminal-power relation using $P_{\text{terminal}} = N_{\text{eq}} V I$, with all power quantities converted to consistent units. Cell resistive losses are represented within the electrical model and remain separate from PCS losses.

2.3 Thermal Model and Temperature Pathways

A single-node thermal model represents the temperature of the equivalent cell. Two pathways are distinguished. In Path A, the external thermal boundary changes cell temperature and therefore electrical properties and feasible power. In Path B, battery current generates heat that changes temperature and feeds back on the electrical model.

The nominal annual calculation applies the Phoenix outdoor-temperature record directly as the external boundary. This exposed-cell configuration defines the thermal conditions of the reference comparison. Additional calculations use constant boundaries of 25, 30, and 35 ℃ to assess the dependence of the results on the imposed thermal environment.

Thermal resistance R_{th}, thermal capacitance C_{th}, and an effective heat-generation coefficient κ were identified from periodic-pulse experiments. The modeled heat source is $\kappa I^2 R_{\text{eq}}$. Reversible entropic heat is excluded because an entropy-coefficient map is not available from the identification data. With current, resistance, and boundary temperature held constant within an interval, the single-node model has the discrete solution:

$$\begin{aligned} \mid T_{k+1} \mid &= a_T T_k + (1 - a_T) \left[T_{\text{amb},k} + R_{\text{th}}\, \kappa I_k^2 R_{\text{eq}}\ (z_k, T_k) \right] \\ \mid a_T \mid &= \exp\left(-\frac{\Delta t_s}{R_{\text{th}}\, C_{\text{th}}} \right), \end{aligned} \tag{8}$$

where, $T_{\text{amb},k}$ is the imposed boundary temperature and Δt_s = 3600Δt is the interval duration in seconds. The identified values are R_{th} = 2.886 K·W^{-1}, C_{th} = 130.508 J·K^{-1}, and κ = 0.5828, giving a thermal time constant of 376.68 s. The coefficient κ is an empirical parameter of the reduced thermal model.

A 55 ℃ dispatch cutoff is adopted from the upper operating-temperature value reported in the manufacturer data [17]. Charging and discharging are disabled for an interval when its initial cell temperature is at or above this value. The temperature state continues to evolve under the external boundary. Because the cutoff is evaluated at interval boundaries, it does not impose a continuous-time upper bound on cell temperature.

Temperature updates the electrical properties and voltage-limited current capability used in the following interval. Path A is evaluated by retaining temperature-dependent electrical behavior while setting the self-heating term in Eq. (8) to zero. The incremental contribution of Path B is evaluated by restoring this term under the same external boundary and dispatch formulation.

2.4 Dispatch Cases and Solution Procedure

The nominal dispatch horizon is 24 hours, divided into 96 intervals of 15 minutes. Initial and terminal SOC are fixed at 0.50 in every optimization block. Cell temperature is propagated continuously between consecutive optimization blocks. Only SOC is constrained to the prescribed block-boundary value. These constraints establish a common terminal-state condition and prevent net SOC transfer between blocks. The nominal 30 MW/140 MWh rating corresponds to a rated duration of 4.67 h and a power-to-energy ratio of 0.214 h^{-1}. Table 2 summarizes the dispatch cases and operating conditions.

Electrical-only dispatch (B1). The baseline model uses a fixed energy capacity of E = 140 MWh and the state equation:

$$z_{k+1} = z_k + \frac{\eta_{\text{ch}} P_{\text{ch},k} \Delta t}{E} - \frac{P_{\text{dis},k} \Delta t}{\eta_{\text{dis}} E} \tag{9}$$

with $\eta_{\text{ch}} = \eta_{\text{dis}}$ = 0.98. Its 0.10–0.90 SOC window represents 112 MWh of usable energy. Dispatch minimizes:

$$\min \left[\sum_k P_{\text{curt},k} \Delta t + \varepsilon \sum_k \left(P_{\text{ch},k} + P_{\text{dis},k} \right) \Delta t \right], \varepsilon = 10^{-7} \tag{10}$$

Table 2. Dispatch cases and reference operating conditions

Case or Parameter	Definition or Value
B0	No storage; PV export subject to the POI limit
B1	Electrical-only optimization with fixed efficiencies, energy capacity, and AC power limits
B2	Nonlinear replay of the B1 schedule, followed by terminal-SOC correction
B3	Coupled dispatch using sequential linearization, followed by terminal-SOC correction and final nonlinear replay
PV rating	230 MW AC
Export-cap ratio	0.85 nominal; 0.70 and 0.95 in sensitivity calculations
BESS rating	30 MW/140 MWh nominal; 15 and 60 MW at fixed 140 MWh in power-sensitivity calculations
SOC range	0.10–0.90
Initial and terminal SOC	0.50 in each optimization block
Dispatch interval	15 min; selected stressed days also evaluated at 5 min
PCS efficiency	0.98 in each direction

Note: B0 denotes the no storage scenario, B1 denotes the electrical-only optimization, B2 denotes the nonlinear physical replay of the B1 schedule with terminal-SOC correction, and B3 denotes the coupled dispatch using sequential linearization. PV = photovoltaic; POI = point of interconnection; BESS = battery energy storage system; AC = alternating current; SOC = state of charge; PCS = power-conversion-system.

The small throughput penalty discourages unnecessary cycling among schedules with equivalent curtailment performance. Eq. (10) is subject to the POI constraints, fixed power limits, SOC bounds, and terminal-SOC condition.

Terminal-corrected nonlinear replay (B2). The B1 commands are evaluated sequentially using the temperature-dependent battery model. At each interval, accepted power is restricted by the BESS rating, PV surplus or export headroom, SOC availability, current and voltage limits, and temperature cutoff. Accepted power determines the current used to advance SOC and temperature.

Because the physical model and B1 use different state-conversion relations, direct replay can end with an SOC different from 0.50. A deterministic terminal correction first scales discharge commands. Charge commands are scaled only when discharge scaling cannot bracket the target terminal SOC. Each trial scaling is assessed through a complete nonlinear replay, including all POI and battery constraints. The terminal-SOC tolerance is 10^{-6} p.u.

B2 denotes the accepted trajectory after this correction. Discharge is adjusted first because reducing discharge does not directly increase curtailment in the same interval, although it can affect subsequent charging capacity. The B1–B2 comparison measures the consequences of executing the simplified-model schedule under the nonlinear representation and a common terminal-SOC condition. Thermal-pathway contributions are evaluated separately through the controlled B3 comparisons.

Coupled dispatch (B3). The sequential-linearization procedure starts from the B1 schedule. Each nonlinear replay supplies local charging and discharging state-conversion coefficients and temperature-dependent feasible power limits. These quantities define the next linear dispatch subproblem. The requested schedule is updated with a damping factor of 0.5 before replay.

A candidate is classified as numerically admissible when no requested power is clipped and the absolute terminal-SOC error is below 5×10^{-4} p.u., after at least three replay evaluations. The procedure evaluates at least five replay states and, when the iteration budget permits, one additional state after first satisfying this criterion. The nominal limit is 10 replay evaluations.

Among admissible candidates, the physically accepted trajectory with the lowest curtailment is retained; terminal-SOC error breaks ties. If no candidate meets the admissibility criterion, selection prioritizes the absence of clipping, then smaller terminal-SOC error, and finally lower curtailment. The selected schedule undergoes the same terminal correction as B2. A final nonlinear replay provides all reported energy and operating metrics. B3 is the solution returned by this sequential procedure, whose numerical behavior is assessed through the initialization and extended-iteration tests in Section 4.4.

For the horizon comparison, the annual calculation is repeated using consecutive 48-hour blocks with the same initial and terminal SOC. The final block covers the remaining 24 hours of the 365-day record. Separate full-model and no-self-heating calculations at constant external boundaries of 25, 30, and 35 ℃ assess the influence of the

imposed thermal environment.

3 Data, Parameter Identification, and Validation

3.1 Solar and Meteorological Data

Solar and meteorological inputs were obtained from the National Solar Radiation Database (NSRDB) Geostationary Operational Environmental Satellite (GOES) Conus Physical Solar Model version 4 for Phoenix, Arizona, at approximately 33.44° N, 112.07° W [18]. The 2023 record contains 105,120 five-minute samples of global horizontal irradiance, direct normal irradiance, diffuse horizontal irradiance, air temperature, wind speed, solar zenith angle, and surface pressure. The five-minute series was retained for temporal-resolution tests. Annual dispatch used 15-minute arithmetic means of available PV power and ambient temperature, giving 35,040 intervals.

Available PV power was calculated using a fixed-tilt, PVWatts-based simplified model [19]. The array was south-facing, with a tilt of 20°, a direct current (DC)-to-AC ratio of 1.3, and an inverter efficiency of 0.98. The power-temperature coefficient was $-0.0035\ ^{\circ}\mathrm{C}^{-1}$, the nominal operating cell temperature was 45 °C, and ground albedo was 0.2. Solar geometry was calculated from the site coordinates and timestamps and checked against the NSRDB zenith-angle series, with the solar-position method documented in Reda and Andreas's work [20]. The annual mean absolute error (MAE) and root-mean-square error (RMSE) between the calculated and database zenith angles were 0.214° and 0.248°, respectively. This comparison checks the solar-geometry calculation rather than validating PV power against plant measurements.

Modeled AC output was limited to the 230 MW inverter rating before applying the POI export constraint. The resulting annual available PV energy was 597.198 GWh, corresponding to an AC capacity factor of 29.64%. This available-power series was used consistently across the dispatch comparisons.

3.2 Battery Data and Parameter Identification

Battery parameters were identified from the public dataset for the A123 ANR26650M1-B cylindrical LFP cell, with a nominal capacity of 2.5 Ah [16]. The dataset includes open-circuit-voltage tests and dynamic profiles at temperatures from −25 °C to 45 °C in 10 °C increments, constant-current/constant-voltage charging tests, and periodic-pulse records for thermal characterization. The Urban Dynamometer Driving Schedule (UDDS) profiles at 25 and 35 °C were excluded from parameter identification and reserved for independent validation.

At each test temperature, low-rate charging and discharging traces were converted to SOC and averaged at matched SOC values to construct $U_{\mathrm{oc}}(z,T)$. Shape-preserving piecewise-cubic interpolation was applied along the SOC axis, with linear interpolation between measured temperatures. The temperature-dependent capacity $Q(T)$ was derived from the same low-rate tests.

Ohmic resistance $R_0(z,T)$ was estimated from voltage changes associated with current transitions in the identification profiles. The estimates were grouped into SOC bins before interpolation. Polarization resistance $R_1(T)$ and its time constant were obtained from post-discharge voltage relaxation, separating the transient polarization response from the instantaneous ohmic response. Thermal resistance, thermal capacitance, and the effective heat-generation coefficient were fitted to the periodic-pulse temperature record. The held-out UDDS profiles were not used in these fitting stages.

Figure 2 shows the electrical-property maps used in dispatch. Table 3 distinguishes experimentally identified parameters, manufacturer-based operating bounds, and system-level assumptions.

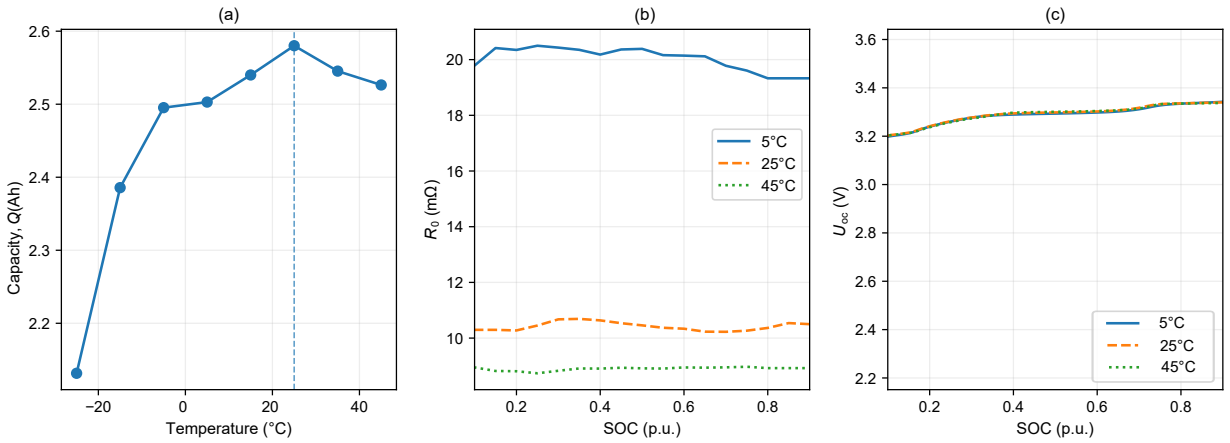


Figure 2. Identified cell properties: (a) capacity, (b) ohmic resistance, and (c) open-circuit voltage

Note: Panels (b) and (c) show 5, 25, and 45 °C [16]. SOC = state of charge.

Table 3. Data sources, model parameters, and operating assumptions

Item	Value or Specification	Source or Basis
Solar and weather record	Phoenix, Arizona; 2023; five-minute resolution; 33.44° N, 112.07° W	NSRDB GOES Conus Physical Solar Model v4 [18]
Annual dispatch input	15-minute means; 35,040 intervals	Aggregated from the five-minute series
PV configuration	230 MW AC; tilt 20°; azimuth 180°; DC-to-AC ratio 1.3	Study assumptions; PVWatts-like model [19]
PV conversion parameters	Inverter efficiency 0.98; power-temperature coefficient $-0.0035\ ^{\circ}\mathrm{C}^{-1}$; nominal operating cell temperature 45 °C	Study assumptions
Ground albedo	0.2	Study assumption
Battery cell	A123 ANR26650M1-B; LFP; nominal capacity 2.5 Ah	Public experimental dataset [16]
Electrical identification range	−25 °C to 45 °C	Experimental dataset [16]
Dispatch SOC range	0.10–0.90	Study operating assumption
Thermal resistance	$2.886\ \mathrm{K\cdot W^{-1}}$	Identified from periodic-pulse data
Thermal capacitance	$130.508\ \mathrm{J\cdot K^{-1}}$	Identified from periodic-pulse data
Effective heat-generation coefficient	0.5828	Identified reduced-model parameter
Charging-current bound	10 A	Adopted from the recommended fast-charge current [17]
Discharging-current bound	50 A	Manufacturer continuous-discharge rating [17]
Terminal-voltage bounds	2.0–3.6 V	Adopted from the manufacturer discharge-characteristic endpoint and recommended charge endpoint [17]
Temperature cutoff	55 °C, evaluated at interval boundaries	Adopted upper operating-temperature value [17]
PCS efficiency	0.98 in each direction	Common assumption across dispatch cases

Note: NSRDB = National Solar Radiation Database; GOES = Geostationary Operational Environmental Satellite; PV = photovoltaic; SOC = state of charge; PCS = power-conversion-system; AC = alternating current; DC = direct current; LFP = lithium iron phosphate.

3.3 Independent Electrothermal Validation

The identified electrothermal model was evaluated against the complete held-out UDDS profiles at 25 and 35 °C. Validation retained the first-order polarization state, whereas annual dispatch used the quasi-steady representation defined in Section 2.2. The two tests therefore assess the identified dynamic cell model at the validation temperatures; they do not constitute a separate validation of the annual model reduction.

Measured and simulated values were compared at the native timestamps. No filtering, resampling, offset correction, initialization trimming, or sample exclusion was applied. The 25 °C profile contained 8,326 samples spanning 1.052–8,440.17 s. The 35 °C profile contained 8,342 samples spanning 1.053–8,440.17 s. Both profiles had a median sampling interval of 1.014 s.

At 25 °C, voltage RMSE and MAE were 0.03193 and 0.02798 V, respectively, while temperature RMSE and MAE were 0.07978 and 0.06037 °C. At 35 °C, voltage RMSE and MAE were 0.06759 and 0.03712 V, and temperature RMSE and MAE were 0.1524 and 0.1132 °C. Figure 3 compares the complete measured and simulated trajectories.

Independent validation was available at 25 and 35 °C, while parameter identification covered −25 to 45 °C. These two domains are distinguished throughout the analysis. The controlled thermal-coupling map remains within the identification range. For annual trajectories above 45 °C, electrical-property lookup values are held at the upper identification boundary, and the effect of this treatment is assessed separately.

3.4 Scenario Design and Mechanism Attribution

The scenarios address three aspects of model adequacy: annual curtailment, the physical feasibility of a prescribed schedule, and the separate contributions of ambient-driven electrical-property variation and current-driven self-

heating.

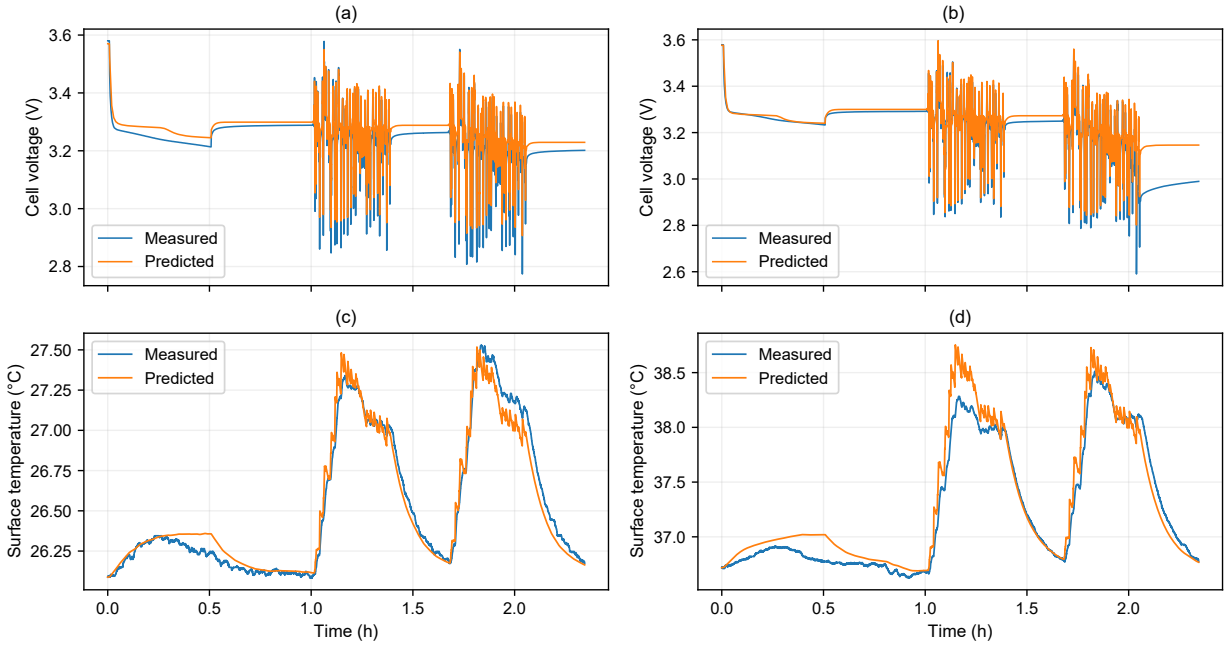


Figure 3. Measured and simulated voltage and temperature for the held-out UDDS profiles at 25 and 35 °C [16]: (a) voltage at 25 °C, (b) voltage at 35 °C, (c) temperature at 25 °C, and (d) temperature at 35 °C

Note: UDDS = Urban Dynamometer Driving Schedule.

Annual operating conditions. The annual screen varied the export-cap ratio over 0.70, 0.85, and 0.95 and BESS power over 15, 30, and 60 MW at a fixed energy rating of 140 MWh. Thermal boundaries comprised the original outdoor-temperature record and versions shifted by −10 and +10 °C. Available PV power was held unchanged in these temperature-shift cases so that the comparison isolated battery-temperature effects from changes in PV conversion.

The +10 °C boundary was treated as an elevated-temperature stress test: approximately 17% of intervals exceeded the 45 °C upper electrical-identification temperature. Separate constant-boundary calculations at 25, 30, and 35 °C assessed temperature dependence within the identification range. These prescribed boundaries represent controlled thermal conditions without specifying an enclosure or heating, ventilation, and air conditioning (HVAC) design.

Thermal-coupling map. The operating days were selected as the union of the eight hottest and eight highest-curtailment days in the 2023 record, giving 16 days. For each selected day, the prescribed B1 command sequence was replayed at constant boundary temperatures of 5, 15, 25, 35, and 40 °C. BESS power remained at 30 MW, while rated energy was set to 140, 70, 35, 20, 10, and 5 MWh. The resulting power-to-energy ratios ranged from 0.214 to 6 h^{-1}.

The 30 temperature–energy combinations were evaluated using paired full-model and no-self-heating replays with identical commands. Recorded quantities included maximum self-heating temperature rise, representative-cell current, charging-current utilization, command clipping, and curtailed energy. Keeping the requested schedule unchanged across the paired replays separated the physical effect of self-heating from changes introduced by re-optimization.

Pathway and property attribution. Controlled B3 comparisons used the nominal annual boundary and the common 25 °C-normalized usable-energy basis. In the fixed-property reference, open-circuit voltage, equivalent resistance, capacity, and power-feasibility calculations used their 25 °C values, and self-heating was disabled. The Path-A case activated temperature-dependent electrical properties and feasibility limits while retaining zero self-heating. The full model then restored the current-driven heat source.

One-at-a-time ablations activated temperature dependence in capacity, equivalent resistance, or open-circuit voltage while retaining fixed-reference power-feasibility calculations. A separate feasibility-only case activated temperature-dependent admissible power and the temperature cutoff while keeping state-conversion properties fixed at 25 °C. These ablations identify individual mechanisms; their effects are not assumed to be additive because the dispatch and physical constraints are nonlinear.

Thermal-parameter sensitivity was evaluated by varying R_{th}, C_{th}, and κ individually by $\pm 20\%$. Capacity-map sensitivity used charge-derived and discharge-derived capacity estimates and shape-preserving piecewise-cubic Hermite interpolation (PCHIP) between temperature points. All capacity variants retained the same 25 °C usable-energy normalization.

Resolution, horizon, and identification-domain checks. The same 16 selected days were evaluated at five-minute resolution to assess sensitivity to the 15-minute annual interval. The annual dispatch was also repeated with 48-hour blocks under the terminal-SOC conditions defined in Section 2.4. An additional identification-domain check separated the Path-A contribution of days containing temperatures above 45 ℃ and repeated the calculation with the thermal state capped at 45 ℃.

Annual energy balance, terminal SOC, and POI feasibility were checked using the physically accepted trajectories. All reported B3 energy metrics were calculated from the final nonlinear replay after terminal-SOC correction.

3.5 Computational Implementation and Numerical Verification

Dispatch calculations were implemented in Python, with linear subproblems solved using the HiGHS linear-programming backend [21]. The same nonlinear replay model and identified parameter maps were used to evaluate B2, B3, and the controlled mechanism comparisons.

For each B3 optimization block, the recorded diagnostics included the number of replay evaluations, satisfaction of the admissibility criterion defined in Section 2.4, linear-subproblem failures, and selection of an earlier replay candidate. Recording candidate selection distinguished the final reported trajectory from the last iteration of the sequential procedure.

Sensitivity to initialization was tested by starting B3 from the terminal-corrected B2 trajectory instead of B1. Blocks that had not met the strict admissibility criterion within 10 replay evaluations were also extended to a maximum of 25 evaluations. Changes in annual curtailment from these tests were used to assess the numerical sensitivity of the dispatch solution and to interpret the scale of the pathway differences reported in Section 4.

4 Results

4.1 Annual Curtailment Recovery and Dispatch Improvement

The reference PV plant produced 597.198 GWh of available AC energy in 2023. With an export-cap ratio of 0.85 and no storage, annual curtailment was 39.601 GWh, equivalent to 6.631% of available generation. Electrical-only dispatch (B1) reduced curtailment to 8.727 GWh, recovering 30.874 GWh of otherwise curtailed PV energy (Table 4).

Table 4. Annual curtailment and recovery for the reference PV–battery system

Case	Curtailment (GWh)	Curtailment (% of Available PV Energy)	Curtailment Avoided Relative to B0 (GWh)
B0	39.601	6.631	—
B1	8.727	1.461	30.874
B2	9.032	1.512	30.569
B3	8.956	1.500	30.645

Note: Differences reported in the text were calculated from unrounded outputs. Avoided curtailment denotes PV energy accepted instead of curtailed, before subsequent storage losses. B0 denotes the no storage scenario, B1 denotes the electrical-only optimization, B2 denotes the nonlinear physical replay of the B1 schedule with terminal state of charge correction, and B3 denotes the coupled dispatch using sequential linearization. PV = photovoltaic; "—" = not applicable.

Direct nonlinear replay of the B1 schedule increased curtailment to 8.948 GWh before terminal-SOC correction. The 221.1 MWh difference from B1 reflected changes in accepted power and state conversion under the physical battery model. The replay also produced a mean absolute daily terminal-SOC error of 0.00338 p.u. and a maximum error of 0.01247 p.u.; 256 daily blocks exceeded 0.001 p.u.

Applying the POI-aware terminal correction gave B2 curtailment of 9.032 GWh. The corrected trajectory contained no charging above available surplus or discharging above export headroom, and its maximum terminal-SOC error was 1.05×10^{-12} p.u. On this common terminal-SOC basis, the B1–B2 curtailment difference was 305.2 MWh/year.

Coupled dispatch (B3) reduced annual curtailment to 8.956 GWh, recovering 30.645 GWh relative to operation without storage. This was 75.78 MWh less curtailment than B2. The improvement resulted from revising the dispatch schedule using the nonlinear battery representation; it is distinct from the individual temperature-pathway effects evaluated in Section 4.2.

4.2 Temperature Pathways and Their Contributions

Figure 4 compares B2 and B3 on 20 July 2023, a hot day with high PV output. Both schedules charged from the midday surplus, but their discharge trajectories differed because B3 updated dispatch using temperature-dependent state conversion and feasible power. At the nominal 30 MW/140 MWh rating, cell temperature closely followed

the imposed ambient boundary. The thermal response on this day was therefore governed primarily by external conditions.

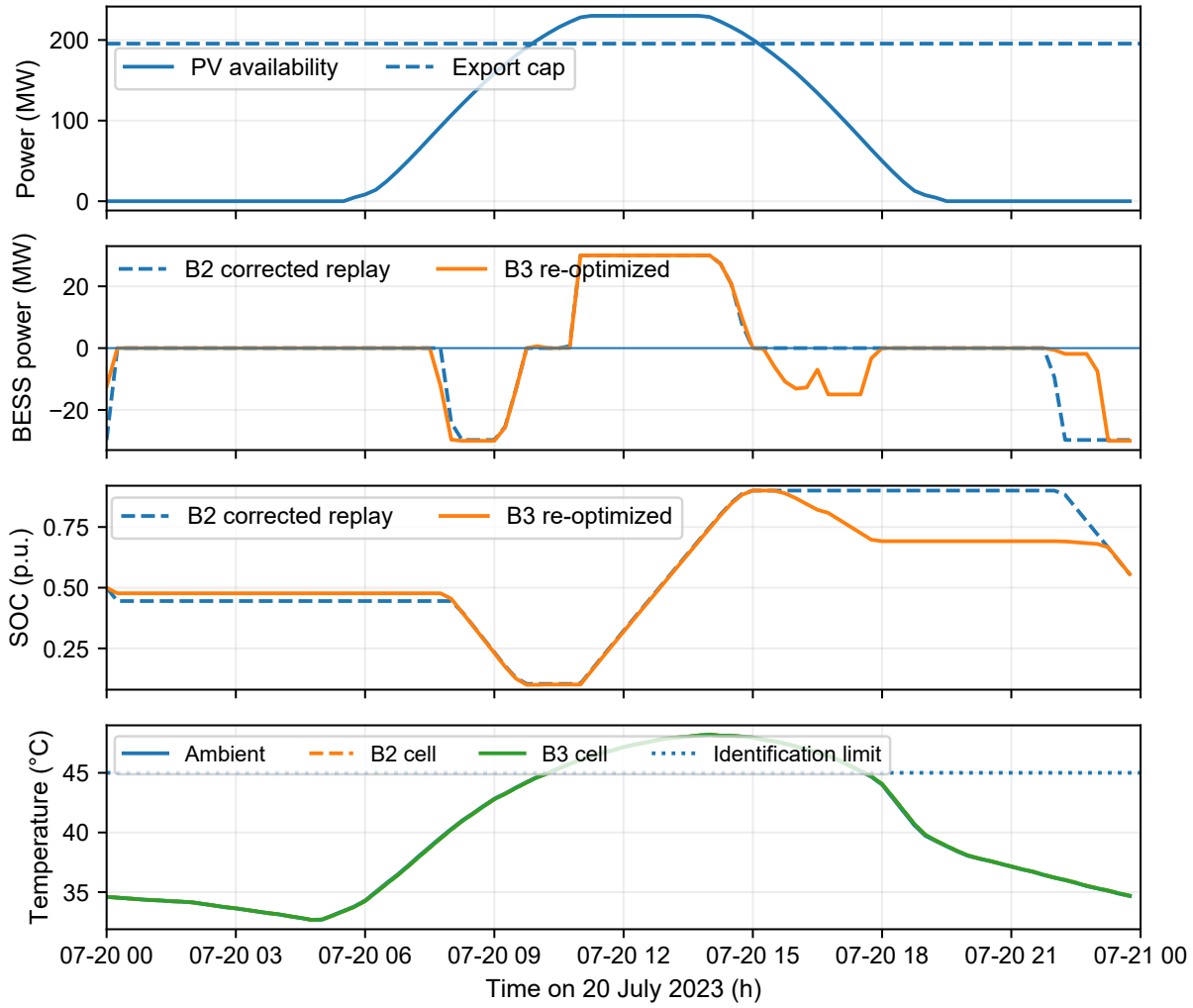


Figure 4. B2 and B3 operation on 20 July 2023

Note: B2 denotes the nonlinear physical replay of the electrical-only optimization schedule with terminal state of charge correction, and B3 denotes the coupled dispatch using sequential linearization. Positive battery power denotes charging. SOC = state of charge; BESS = battery energy storage system; PV = photovoltaic.

The controlled annual comparisons separated the environmental-property pathway from current-driven self-heating (Figure 5). With electrical properties fixed at 25 ℃, terminal-corrected B2 curtailment was 8.729 GWh, 1.890 MWh above B1. This small replay difference contrasted with the 305.2 MWh difference under the nominal temperature-dependent model.

Re-optimization with the fixed-25 ℃ physical representation gave B3 curtailment of 8.690 GWh. Activating ambient-driven electrical-property variation while suppressing self-heating increased curtailment by 265.6 MWh/year, to 8.956 GWh. Restoring self-heating produced a raw additional difference of 0.128 MWh/year. This difference was smaller than the changes obtained in the numerical sensitivity tests reported in Section 4.4 and was not resolved as a separate annual dispatch effect.

The property-level ablations identified temperature-dependent capacity as the main contributor to Path A. Activating $Q(T)$ alone increased curtailment by 263.3 MWh/year relative to the fixed-property reference. The corresponding changes were −7.733 MWh/year for equivalent resistance, +0.269 MWh/year for open-circuit voltage, and +3.089 MWh/year for temperature-dependent power feasibility. These are separate one-at-a-time comparisons rather than additive components of the complete Path-A result.

The pathway ordering persisted when thermal resistance, thermal capacitance, and the heat-generation coefficient were varied individually by ±20%. Across the six perturbations, full-model minus no-self-heating differences ranged from approximately 0.15 to 0.23 MWh/year. These remained below the observed annual solution sensitivity.

Capacity-map variants affected the magnitude of Path A more strongly. On the same 25 ℃-normalized usable-

energy basis, charge-derived, discharge-derived, and PCHIP capacity maps gave Path-A effects of 295.5, 233.4, and 261.9 MWh/year, respectively, compared with 265.6 MWh/year for the nominal mean-linear map. The 233.4–295.5 MWh/year range describes sensitivity to these empirical capacity representations.

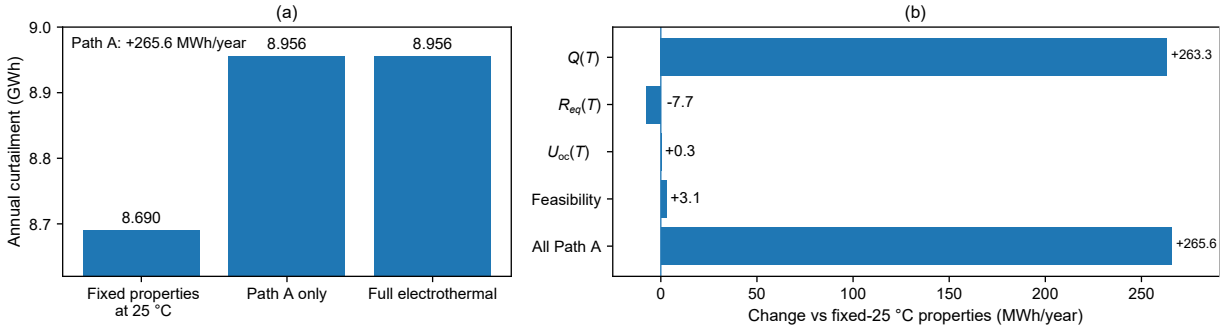


Figure 5. Annual curtailment changes from (a) temperature-pathway activation and (b) individual property ablations

Note: Values are rounded; differences were calculated from unrounded outputs. $Q(T)$ = capacity; $R_{eq}(T)$ = quasi-steady resistance; $U_{oc}(T)$ = open-circuit voltage.

In the nominal full-model calculation, annual cell resistive loss was 221.7 MWh and PCS loss was 1,209 MWh. Charge-plus-discharge throughput was 59.86 GWh, and discharged energy was 29.21 GWh, equivalent to 260.8 full discharge cycles when divided by the 112 MWh reference usable-energy window. These cycles describe operational throughput rather than predicted battery life.

Maximum cell temperature was 48.20 °C, and the 95th percentile was 41.46 °C. Temperature exceeded the 45 °C identification boundary for 102.5 h, or 1.17% of the year, but did not reach the 55 °C dispatch cutoff. Maximum representative-cell current was 0.580 A, equivalent in magnitude to 5.8% of the adopted charging-current bound.

Twenty-three days contained temperatures above 45 °C. Their aggregate contribution to Path A was 46.62 MWh, or 17.55% of the annual effect; the remaining days contributed 219.0 MWh, or 82.45%. These are day-level contributions, not an attribution solely to above-boundary intervals. Capping the thermal state at 45 °C changed Path A by 3.1×10^{-5} MWh/year. Under the adopted boundary-clamped electrical-property lookup, the two treatments therefore produced effectively the same annual attribution.

4.3 Thermal Response and Schedule Feasibility Across Operating Conditions

Figure 6 summarizes the 30 temperature–energy combinations evaluated over the 16 selected days. Paired full-model and no-self-heating replays used identical B1 commands, allowing changes in thermal response and accepted power to be assessed without re-optimization.

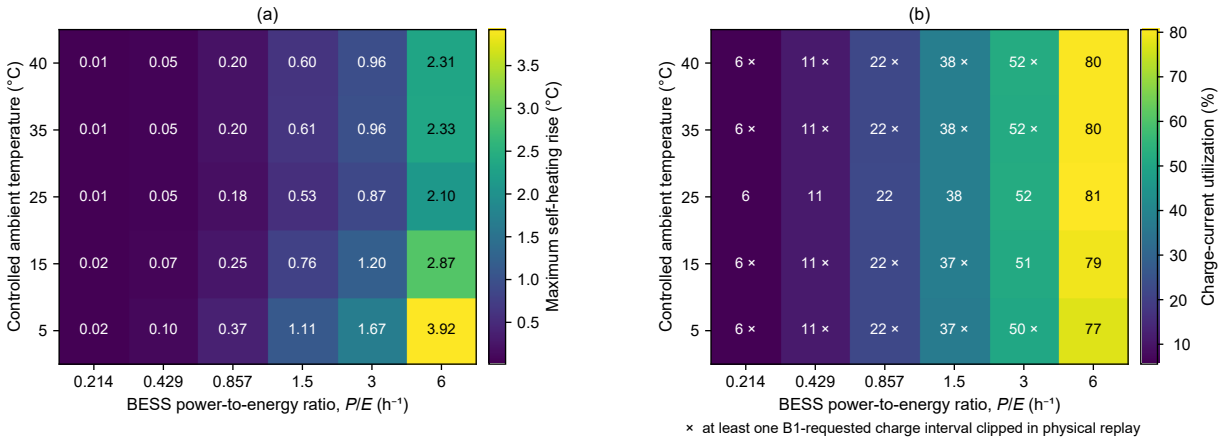


Figure 6. Thermal-coupling map for the 16 selected days: (a) maximum self-heating rise and (b) maximum charging-current utilization

Note: Crosses indicate clipped charging requests. Utilization values are rounded to whole percentages. BESS = battery energy storage system.

Self-heating increased with the BESS power-to-energy ratio. At the nominal ratio of 0.214 h^{-1}, the largest temperature rise above the controlled boundary was approximately 0.025 °C. Across the map, the maximum rise was

3.92 ℃, charging-current utilization reached 80.6% of the adopted 10 A bound, and maximum cell temperature was 42.31 ℃. All mapped thermal trajectories remained within the electrical-identification domain.

Nineteen of the 30 combinations contained at least one requested charging interval that was clipped during physical replay. Nevertheless, the maximum absolute curtailment difference between paired full-model and no-self-heating replays, aggregated over the selected days, was 0.0734 MWh. The mapped results thus distinguished three responses: temperature rise, command feasibility, and recovered energy. A schedule could encounter physical clipping while the additional curtailment caused specifically by self-heating remained small.

The annual temperature-shift cases showed how the broader physical-replay difference changed under hotter external conditions. At the nominal 30 MW rating and original ambient boundary, the uncorrected replay difference from B1 was 221.1 MWh/year. With the ambient record shifted upward by 10 ℃ and α = 0.85, the corresponding differences were 0.859, 1.788, and 1.949 GWh for BESS power ratings of 15, 30, and 60 MW, respectively.

The shifted external boundary reached 58.2 ℃. Battery power was disabled whenever the interval-start temperature reached the 55 ℃ cutoff, while the thermal state continued to respond to the hotter boundary. These results describe an elevated-temperature stress test with boundary-clamped electrical properties. They quantify the behavior of the specified model under that boundary rather than independently validated high-temperature battery performance.

4.4 Numerical Sensitivity and Boundary Dependence

The temporal-resolution comparison repeated the 16 selected days at five-minute resolution. Aggregate B3 curtailment was 828.2 MWh with 15-minute inputs and 830.6 MWh with five-minute inputs. The difference calculated from unrounded outputs was 2.427 MWh, or 0.293%. The largest absolute daily difference was 0.506 MWh, and the largest difference in daily peak temperature was 0.027 ℃. These results characterize resolution sensitivity for the selected hot or high-curtailment days.

The nominal B3 procedure met the admissibility criterion defined in Section 2.4 within 10 replay evaluations for 205 of 365 daily blocks (56.2%). The mean number of replay evaluations was 8.14. In total, 193 blocks reached the 10-evaluation limit, no linear subproblem failed, and an earlier replay candidate was retained in 235 blocks. After terminal correction, the maximum terminal-SOC error was 9.6×10^{-8} p.u.

Starting B3 from the B2 trajectory changed annual curtailment by +0.809 MWh (+0.0090%). Extending the 160 blocks that had not met the admissibility criterion to a maximum of 25 replay evaluations changed the reconstructed annual result by -1.886 MWh (-0.0211%). Of these blocks, 100 still did not meet the same criterion.

Using 48-hour blocks gave full-model B3 curtailment of 8.957 GWh, 1.12 MWh or 0.0125% above the 24-hour result. The fixed-25 ℃ and Path-A-only results were 8.688 and 8.957 GWh, respectively, giving a Path-A effect of 268.8 MWh/year. The raw full-model minus no-self-heating difference was 0.137 MWh/year. The longer horizon retained the same pathway ordering.

Constant external boundaries of 25, 30, and 35 ℃ gave annual full-model curtailment of 8.691, 8.819, and 8.954 GWh, respectively. The paired full-model minus no-self-heating differences were 0.213, 0.390, and 0.111 MWh/year. Peak self-heating above these boundaries was approximately 0.012–0.014 ℃. At the nominal storage rating, changing the imposed temperature altered annual curtailment, while the incremental self-heating effect remained below the observed numerical sensitivity scale.

5 Discussion

5.1 Environmental Temperature as the Main Dispatch-Level Mechanism

The controlled comparisons identify ambient-driven electrical-property variation as the main temperature-related influence on annual curtailment in the reference system. Under the exposed-cell boundary, Path A increased curtailment by 265.6 MWh/year relative to the fixed-25 modeling reference. Capacity variation accounted for most of this response in the one-at-a-time ablations, and alternative capacity maps retained the pathway ordering while changing its magnitude.

This finding connects established electrothermal battery modeling to the plant-level recovery objective. Earlier work has incorporated electrical–thermal coupling into charging control, power estimation, hybrid-system scheduling, and ageing-aware control [10–15]. Here, the dominant dispatch effect arose through temperature-dependent state conversion before current-driven heating became important. Consequently, low representative-cell current did not establish the adequacy of fixed-property dispatch.

The capacity contribution must be interpreted within the SOC and capacity formulation in Eq. (6). The reported $Q(T)$ ablation measures the influence of that empirical map on the modeled dispatch trajectory. It does not establish a universal capacity penalty for LFP storage. Its magnitude depends on the cell data, energy normalization, imposed thermal boundary, and operating schedule.

5.2 Thermal Response, Physical Feasibility, and Energy Recovery

The operating map shows why model detail should be selected for the quantity being assessed. Increasing the power-to-energy ratio raised self-heating and charging-current utilization. Physical replay clipped charging requests in 19 combinations, yet paired full-model and no-self-heating replays differed by at most 0.0734 MWh over the selected days. Command clipping therefore cannot be attributed to self-heating alone: the full electrical, SOC, and power constraints also determine accepted operation.

Temperature remains relevant to objectives beyond recovered energy. Ageing-aware operation and thermal-management studies address degradation, temperature distribution, cooling, and system efficiency [22–26]. A thermal response that has little effect on curtailment can still matter for these quantities. The present results assess curtailment recovery and schedule feasibility; they do not substitute for pack-level thermal or lifetime assessment.

5.3 Implications for Photovoltaic–Battery Dispatch Modeling

The matched-energy comparisons provide two complementary checks. First, B1–B2 measures the consequences of physically executing a simplified-model schedule with a common terminal SOC. Second, B2–B3 measures the improvement obtained by revising that schedule using the physical representation. In the reference case, these differences were 305.2 and 75.78 MWh/year, respectively.

The fixed-25 ℃ B2 result was only 1.890 MWh above B1, compared with 305.2 MWh under temperature-dependent operation. On the common reference usable-energy basis, this comparison identifies temperature-dependent state conversion as an important source of the replay difference. Re-optimization recovered part of the resulting loss in performance, but the B2–B3 improvement should not be interpreted as the contribution of self-heating. The controlled pathway comparisons show that these are separate quantities.

These results support a model-assessment sequence based on energy normalization, terminal-SOC consistency, physical replay, and re-optimization. A simplified model can first be evaluated against the accepted power trajectory and resulting curtailment. Coupled dispatch is justified when those differences affect the study objective. This approach makes the required model detail depend on the operational consequence rather than a prescribed temperature or C-rate threshold.

Avoided curtailment also requires careful interpretation. It measures PV energy accepted by storage instead of being curtailed, whereas subsequent cell and PCS losses determine how much of that energy is delivered later. Assessments of renewable-energy utilization should retain this distinction.

5.4 Applicability and Modeling Boundaries

The reference system uses one LFP cell type and an energy-equivalent population of identical cells. It represents aggregate state conversion and current-dependent heating without resolving pack topology, cell imbalance, interconnect resistance, spatial temperature gradients, coolant distribution, or local hot spots. These effects require pack-specific electrical and thermal information.

The 265.6 MWh/year Path-A result applies to the exposed-cell boundary used in the nominal annual case. Constant-boundary calculations provide a comparison at controlled temperatures, but they do not represent the energy consumption or dynamics of a specific HVAC system. Reversible entropic heat and cooling-system demand are outside the formulation.

The identification range and validation temperatures also define the interpretation of the results. Electrical parameters were identified up to 45 ℃, while independent dynamic validation used 25 and 35 ℃ profiles. The nominal annual trajectory exceeded 45 ℃ for 102.5 h, with electrical lookup values held at that boundary. The thermal-state cap test produced almost the same Path-A attribution under this lookup rule; it does not independently validate electrical properties above 45 ℃. The +10 ℃ annual cases are therefore model-based boundary stress tests.

Battery degradation was excluded from the dispatch objective to separate the temperature mechanisms affecting curtailment. Reported throughput describes operating intensity, not lifetime. Likewise, the five-minute comparison and thermal-coupling map cover the 16 selected days, while the annual conclusions derive from the full-year simulations.

B3 is a sequentially computed solution without a certified global optimality bound. The tested initialization and iteration variants changed annual curtailment by up to 1.886 MWh. This observed sensitivity was smaller than the Path-A effect but larger than the reported self-heating differences.

6 Conclusions

This study examined temperature effects on battery dispatch for PV curtailment recovery under a hard export limit. The analysis combined an experimentally parameterized LFP cell model with annual PV–battery dispatch and separated ambient-driven electrical-property variation from current-driven self-heating.

For the 230 MW PV plant with a 195.5 MW export limit and a 30 MW/140 MWh BESS, electrical-only dispatch reduced annual curtailment from 39.601 to 8.727 GWh. Executing that schedule through the nonlinear model and

correcting terminal SOC gave 9.032 GWh. Coupled dispatch reduced curtailment to 8.956 GWh, an improvement of 75.78 MWh/year over the corrected replay.

On a common 112 MWh reference usable-energy basis, ambient-driven electrical-property variation increased curtailment by 265.6 MWh/year relative to fixed-25 ℃ operation under the exposed-cell boundary. Temperature-dependent capacity was the principal mechanism, with alternative empirical maps giving effects of 233.4–295.5 MWh/year. Current-driven self-heating produced annual differences below the observed numerical sensitivity scale at the nominal rating.

The operating map further separated thermal response from energy-recovery consequences. Self-heating reached 3.92 ℃ and charging-current utilization reached 80.6%, while 19 of 30 combinations contained clipped charging requests. The largest paired full-model versus no-self-heating curtailment difference over the selected days was 0.0734 MWh.

The contribution is a model-assessment framework that connects temperature-dependent state conversion and physical feasibility to recoverable PV energy. Common reference energy, consistent terminal SOC, and nonlinear replay are essential to that comparison. The required electrothermal detail follows from its effect on accepted operation and the dispatch objective; weak self-heating alone does not justify a fixed-property battery model.

Author Contributions

Conceptualization, A.M.J. and I.S.B.; methodology, A.M.J.; software, A.M.J.; validation, A.M.J.; formal analysis, A.M.J.; investigation, A.M.J.; data curation, A.M.J.; writing—original draft preparation, A.M.J.; writing—review and editing, A.M.J. and I.S.B.; visualization, A.M.J.; supervision, I.S.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

Solar and meteorological inputs were obtained from NSRDB [18], and the battery measurements were obtained from Mendeley Data [16]. The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During manuscript preparation, the authors used OpenAI ChatGPT to assist with language editing and consistency checking. The authors reviewed and revised the resulting text and take full responsibility for the content of the article.

References

[1] B. Frew, B. Sergi, P. Denholm, W. Cole, N. Gates, D. Levie, and R. Margolis, "The curtailment paradox in the transition to high solar power systems," *Joule*, vol. 5, no. 5, pp. 1143–1167, 2021. https://doi.org/10.1016/j.joule.2021.03.021

[2] E. S. Chatzistylianos, G. N. Psarros, and S. A. Papathanassiou, "Export constraints applicable to renewable generation to enhance grid hosting capacity," *Energies*, vol. 17, no. 11, p. 2588, 2024. https://doi.org/10.3390/en17112588

[3] N. DiOrio, P. Denholm, and W. B. Hobbs, "A model for evaluating the configuration and dispatch of PV plus battery power plants," *Appl. Energy*, vol. 262, p. 114465, 2020. https://doi.org/10.1016/j.apenergy.2019.114465

[4] M. Bahloul, D. Horan, and S. K. Khadem, "Energy storage sizing analysis and its viability for PV power plant clipping losses minimisation," *Electr. Power Syst. Res.*, vol. 225, p. 109837, 2023. https://doi.org/10.1016/j.epsr.2023.109837

[5] M. Bahloul, M. Daoud, and S. K. Khadem, "Optimal dispatch of battery energy storage for multi-service provision in a collocated PV power plant considering battery ageing," *Energy*, vol. 293, p. 130744, 2024. https://doi.org/10.1016/j.energy.2024.130744

[6] A. Castillejo-Cuberos, J. M. Cardemil, and R. Escobar, "Techno-economic assessment of photovoltaic plants considering high temporal resolution and non-linear dynamics of battery storage," *Appl. Energy*, vol. 334, p. 120712, 2023. https://doi.org/10.1016/j.apenergy.2023.120712

[7] W. Gorman, C. Crespo Montañés, A. D. Mills, J. H. Kim, D. Millstein, and R. H. Wiser, "Are coupled renewable-battery power plants more valuable than independently sited installations?" *Energy Econ.*, vol. 107, p. 105832, 2022. https://doi.org/10.1016/j.eneco.2022.105832

[8] R. Zhu, K. Das, P. E. Sørensen, and A. D. Hansen, "A review on energy management system for grid-connected utility-scale renewable hybrid power plants," *WIREs Energy Environ.*, vol. 14, no. 1, p. e70004, 2025. https://doi.org/10.1002/wene.70004

[9] L. Mattia, H. Beiranvand, W. Zamboni, and M. Liserre, "Lithium-ion battery thermal modelling and characterisation: A comprehensive review," *J. Energy Storage*, vol. 129, p. 117114, 2025. https://doi.org/10.1016/j.est.2025.117114

[10] A. K. de Souza, G. Plett, and M. S. Trimboli, "Lithium-ion battery charging control using a coupled electro-thermal model and model predictive control," in *2020 IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LA, USA, 2020, pp. 3534–3539. https://doi.org/10.1109/APEC39645.2020.9124431

[11] M. A. Xavier, A. K. de Souza, and M. S. Trimboli, "An LPV-MPC inspired battery SOP estimation algorithm using a coupled electro-thermal model," in *2021 American Control Conference (ACC)*, New Orleans, LA, USA, 2021, pp. 4421–4426. https://doi.org/10.23919/ACC50511.2021.9483433

[12] T. Mishra, A. Pandey, and M. R. Almassalkhi, "Predictive optimization of hybrid energy systems with temperature dependency," *Electr. Power Syst. Res.*, vol. 235, p. 110767, 2024. https://doi.org/10.1016/j.epsr.2024.110767

[13] Y. Wu, C. Li, X. Li, J. Liu, W. Yang, Z. Cai, and W. Jiang, "Battery energy storage system scheduling optimization based on temperature difference mitigation within module," *J. Energy Storage*, vol. 140, p. 119159, 2025. https://doi.org/10.1016/j.est.2025.119159

[14] X. Dorronsoro, R. de Castro, J. Varela Barreras, E. Garayalde, and U. Iraola, "Battery aging-aware adaptive model predictive control based on coupled semi-empirical electro-thermal and aging models," *Appl. Energy*, vol. 401, p. 126494, 2025. https://doi.org/10.1016/j.apenergy.2025.126494

[15] Z. Gao, C. S. Chin, W. L. Woo, and J. Jia, "Integrated equivalent circuit and thermal model for simulation of temperature-dependent $LiFePO_4$ battery in actual embedded application," *Energies*, vol. 10, no. 1, p. 85, 2017. https://doi.org/10.3390/en10010085

[16] A. K. de Souza, "Lithium-ion battery OCV and dynamic test data of a $LiFePO_4$ cylindrical cell," *Mendeley Data*, 2021. https://doi.org/10.17632/p8kf893yv3.1

[17] "Nanophosphate high power lithium ion cell ANR26650M1-B," A123 Systems, Inc., Technical Data MD100113-02, 2012. https://e2e.ti.com/cfs-file/__key/communityserver-discussions-components-files/196/ANR-A123-26650.pdf

[18] National Laboratory of the Rockies, "NSRDB GOES Conus physical solar model v4 download API," 2026. https://developer.nlr.gov/docs/solar/nsrdb/nsrdb-GOES-conus-v4-0-0-download/

[19] A. P. Dobos, "PVWatts version 5 manual," National Renewable Energy Laboratory, NREL/TP-6A20-62641, 2014. https://doi.org/10.2172/1158421

[20] I. Reda and A. Andreas, "Solar position algorithm for solar radiation applications," *Sol. Energy*, vol. 76, no. 5, pp. 577–589, 2004. https://doi.org/10.1016/j.solener.2003.12.003

[21] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Math. Program. Comput.*, vol. 10, no. 1, pp. 119–142, 2018. https://doi.org/10.1007/s12532-017-0130-5

[22] N. Collath, B. Tepe, S. Englberger, A. Jossen, and H. C. Hesse, "Aging aware operation of lithium-ion battery energy storage systems: A review," *J. Energy Storage*, vol. 55, p. 105634, 2022. https://doi.org/10.1016/j.est.2022.105634

[23] J. Zhao, H. Li, L. Li, and D. Yuan, "HVC-NSGA-III with thermal–electrochemical degradation coupling for four-objective day-ahead BESS dispatch and SOH-adaptive knee-point selection," *Batteries*, vol. 12, no. 4, p. 140, 2026. https://doi.org/10.3390/batteries12040140

[24] S. Yang, Y. Liao, T. Lin, C. Zhang, and X. Ma, "Thermal management of energy storage systems based on integrated energy system optimisation scheduling," *Appl. Therm. Eng.*, vol. 298, p. 130953, 2026. https://doi.org/10.1016/j.applthermaleng.2026.130953

[25] W. Wang, B. Sun, R. Xiong, E. Dahlquist, H. Li, and Q. Sun, "A prediction-based thermal management strategy for stationary battery energy storage systems," *Green Energy Intell. Transp.*, p. 100468, 2026. https://doi.org/10.1016/j.geits.2026.100468

[26] M. Schimpe, M. Naumann, N. Truong, H. C. Hesse, S. Santhanagopalan, A. Saxon, and A. Jossen, "Energy efficiency evaluation of a stationary lithium-ion battery container storage system via electro-thermal modeling and detailed component analysis," *Appl. Energy*, vol. 210, pp. 211–229, 2018. https://doi.org/10.1016/j.apenergy.2017.10.129